MATRIX
Obstacle Avoiding Robot

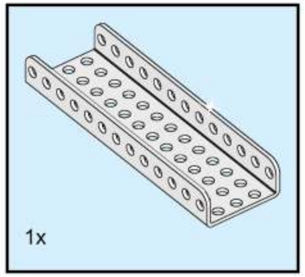
貝登堡智能
KK INTELLIGENT TECHNOLOGY INC.

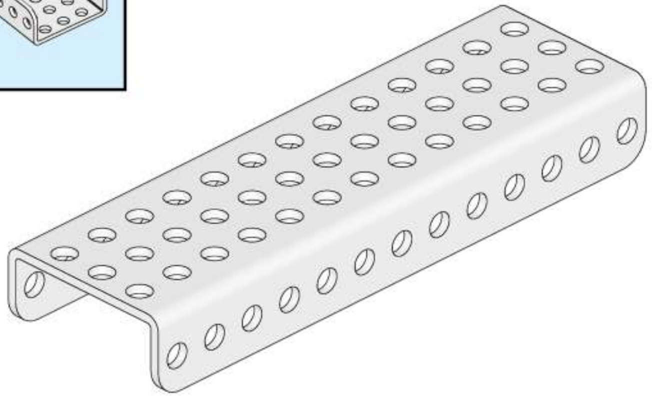
Learning objectives

- Assembling an Obstacle Avoiding Robot with MATRIX Parts
- Understanding Laser Sensor Commands
- Programming an Obstacle Avoidance Robot

Assembly Basic Car

1





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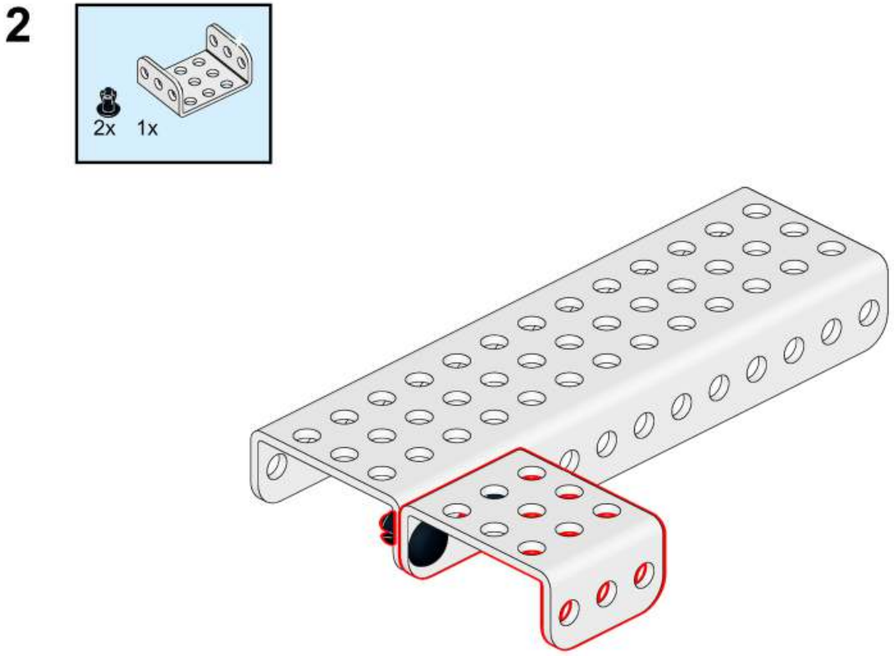

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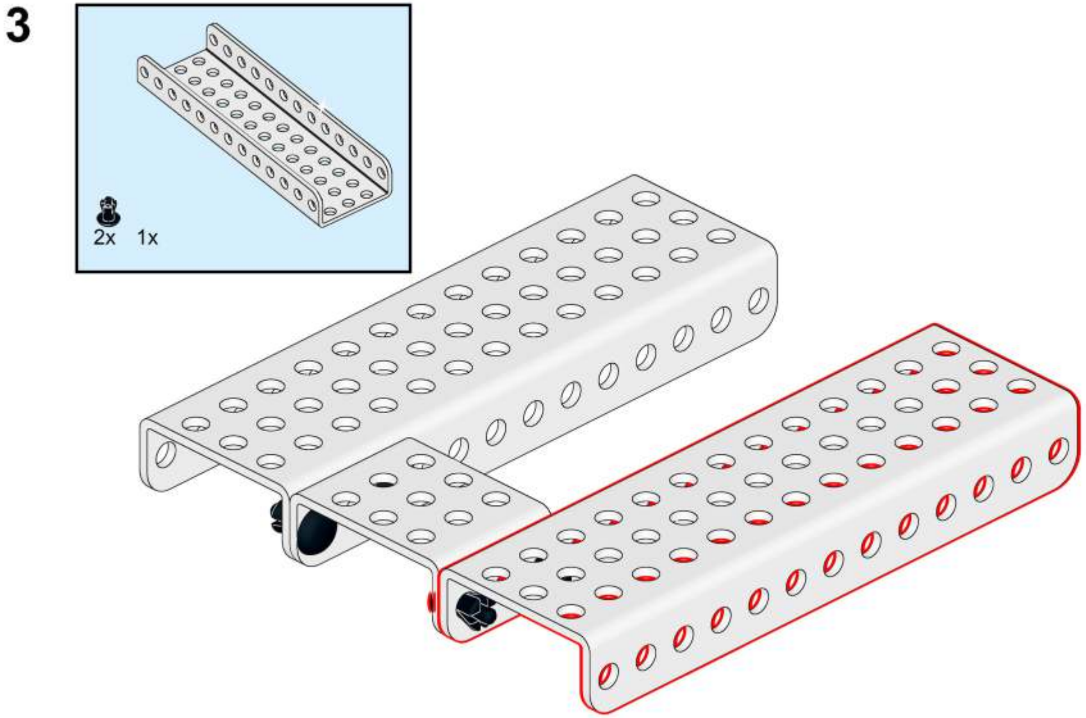

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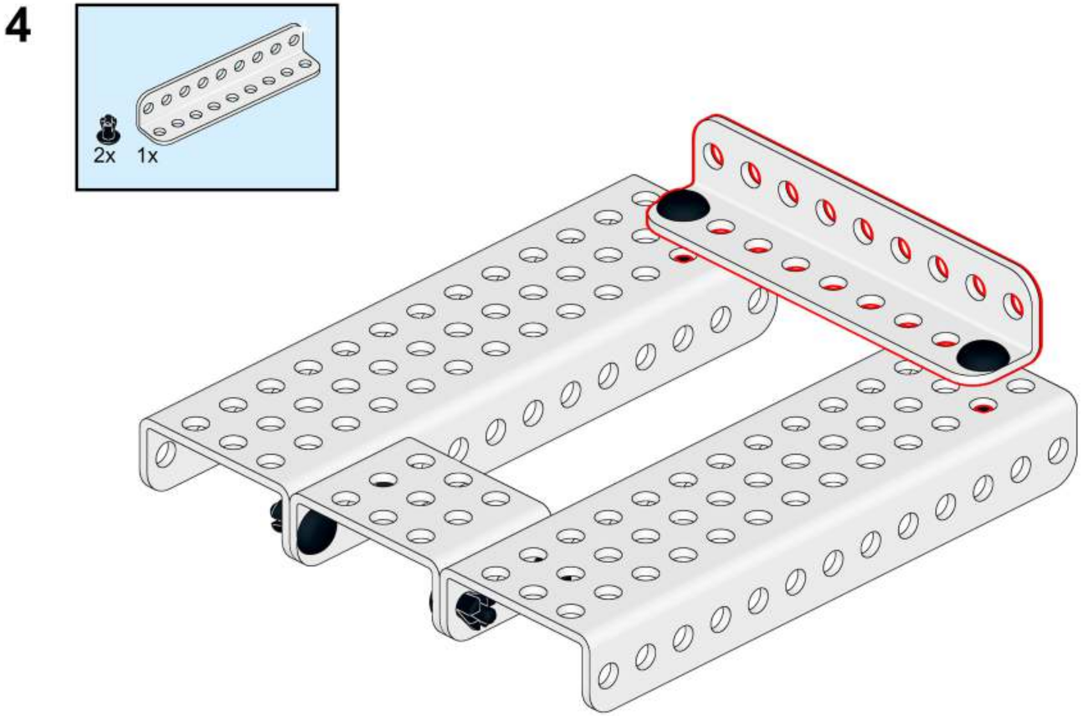
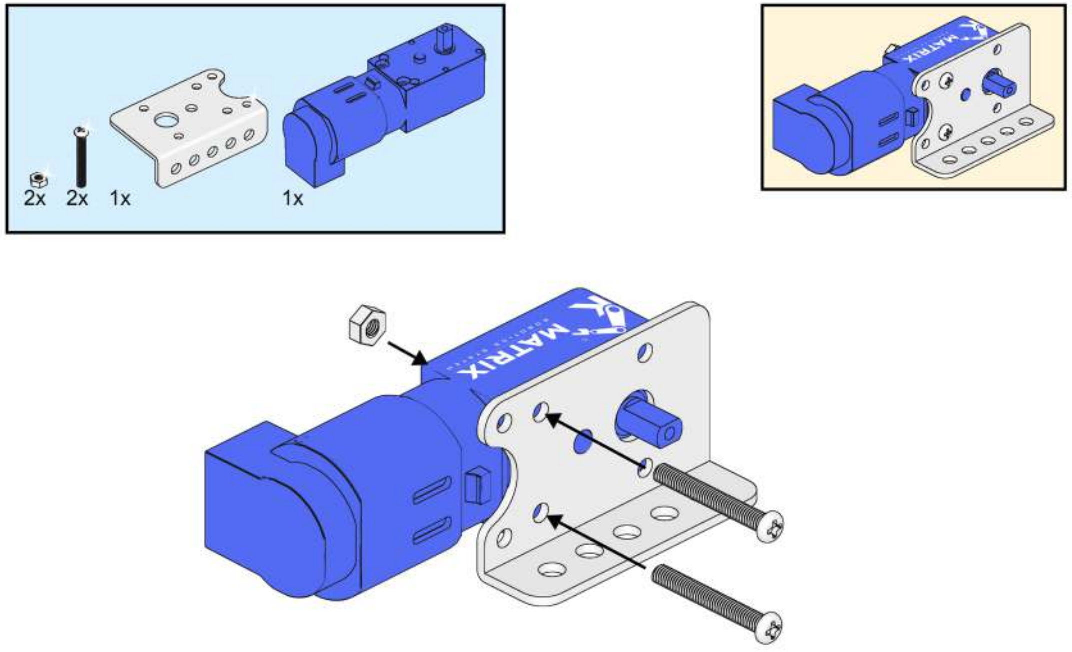
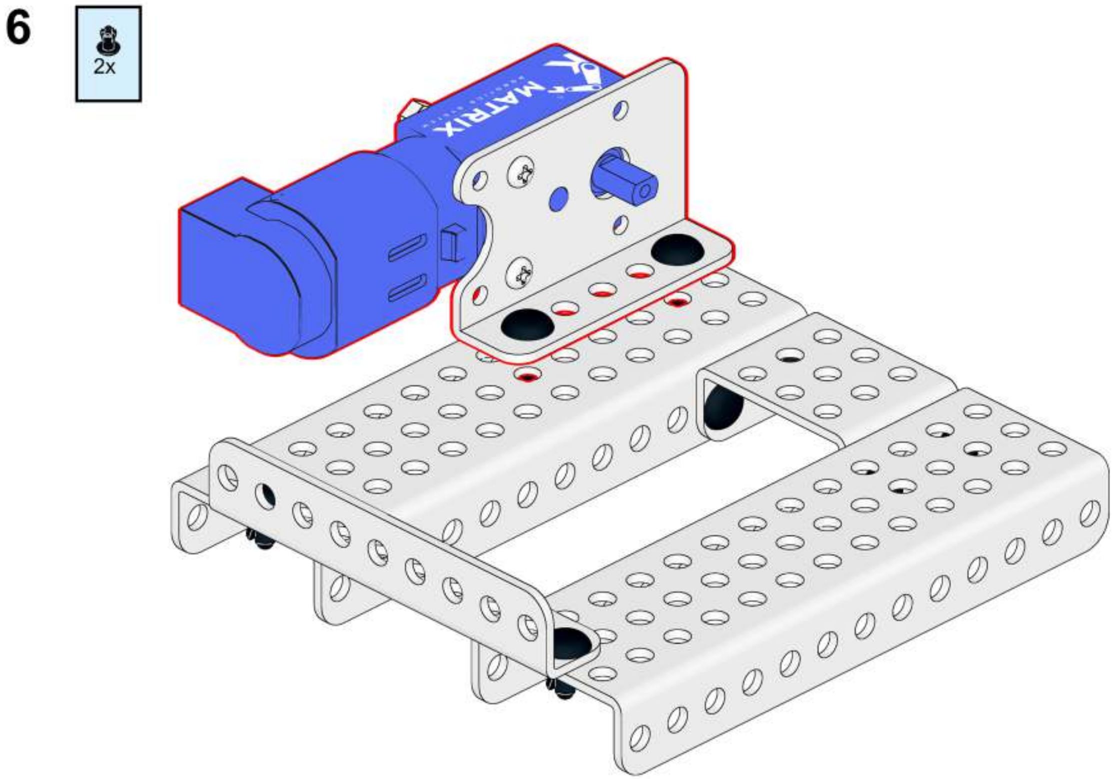


Assembly Basic Car

5

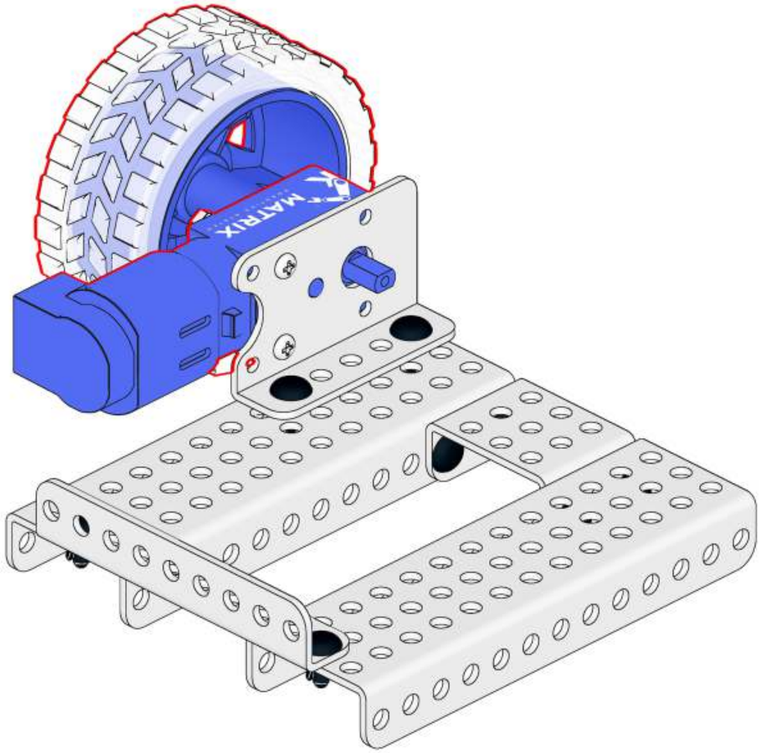


Assembly Basic Car



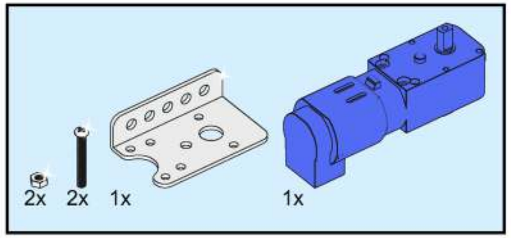
Assembly Basic Car

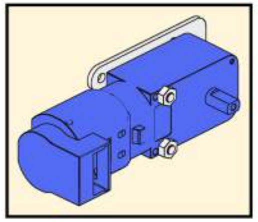
7

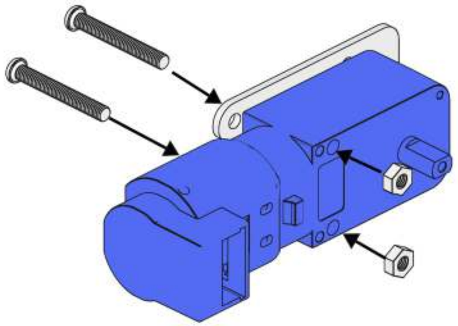


Assembly Basic Car

8







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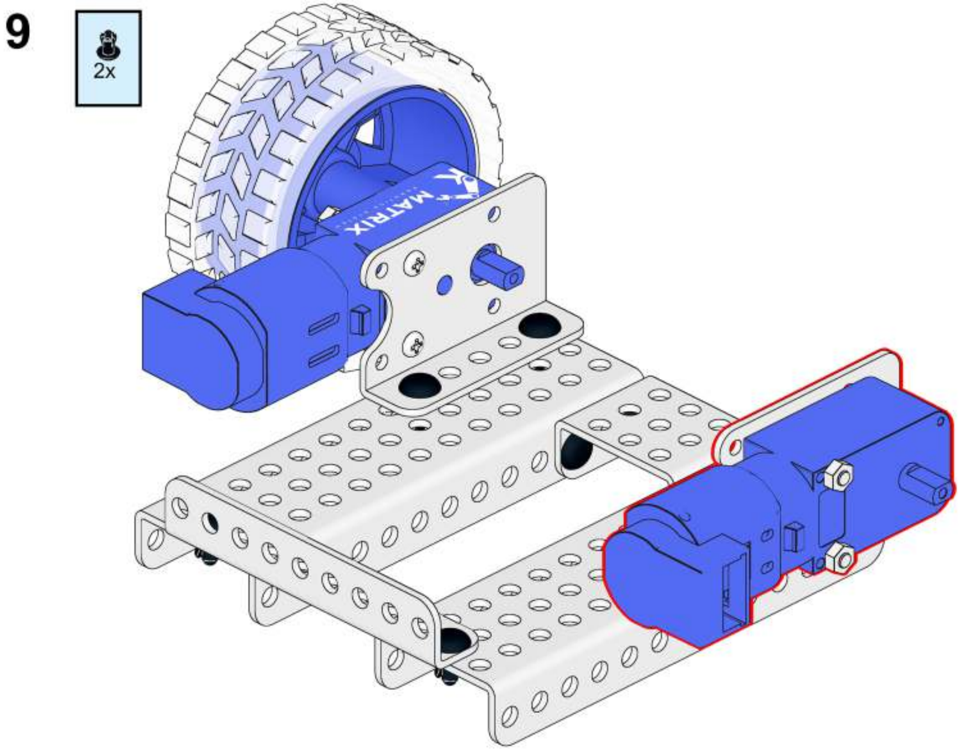
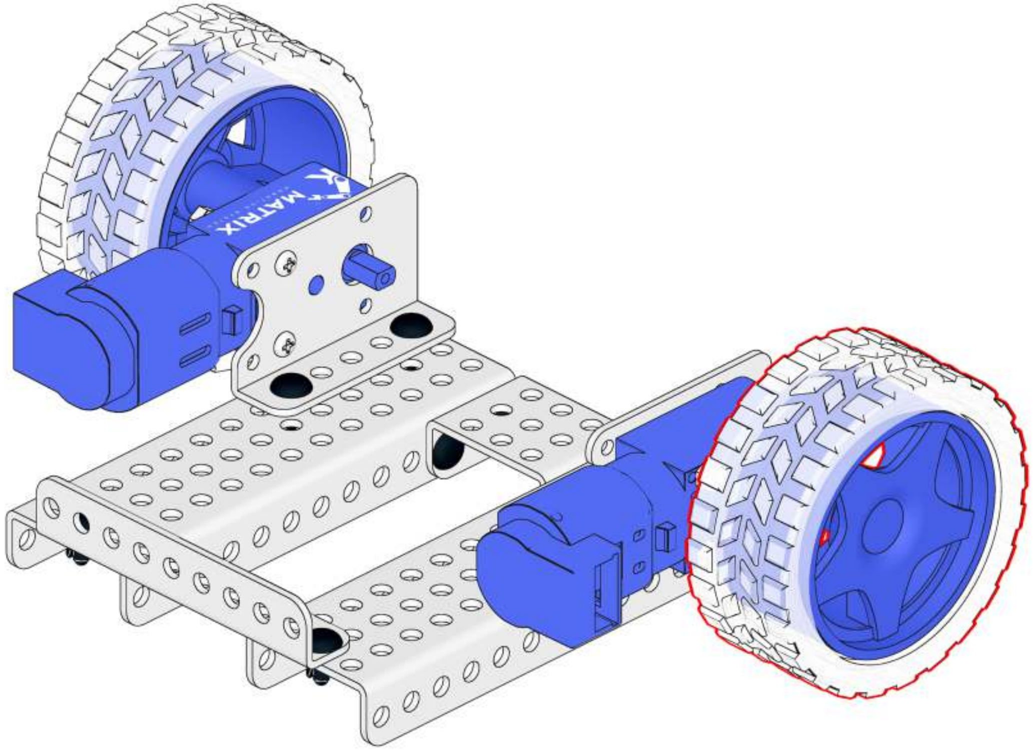
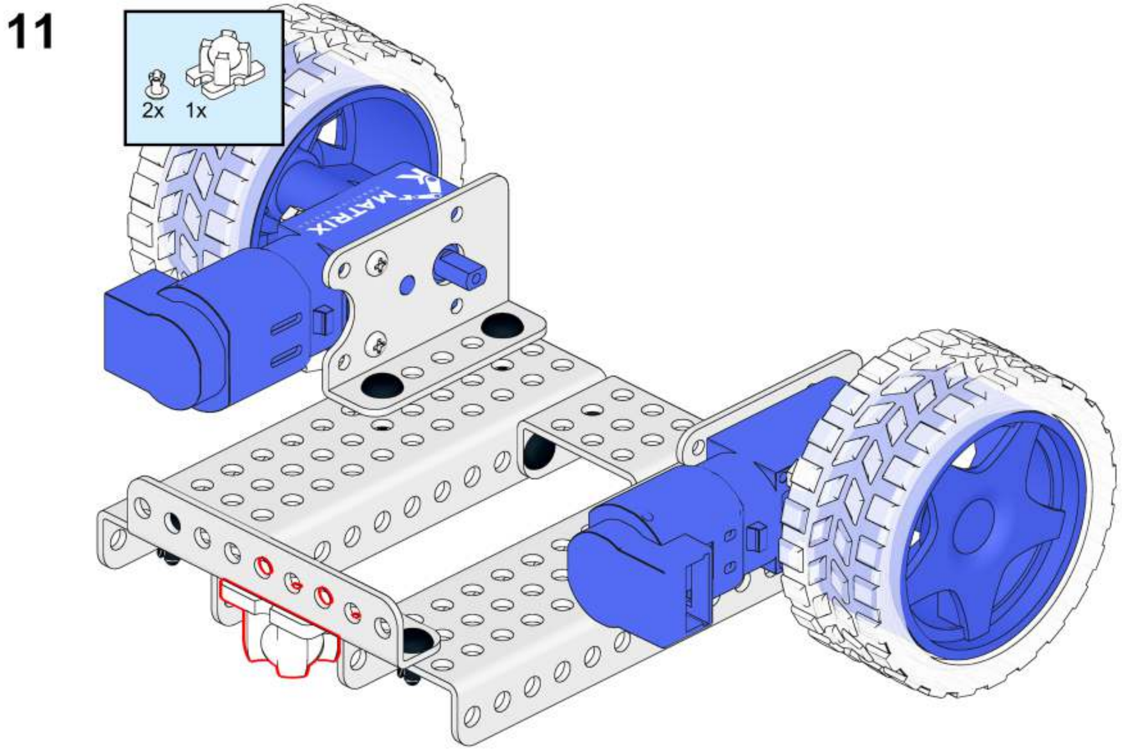


Assembly Basic Car

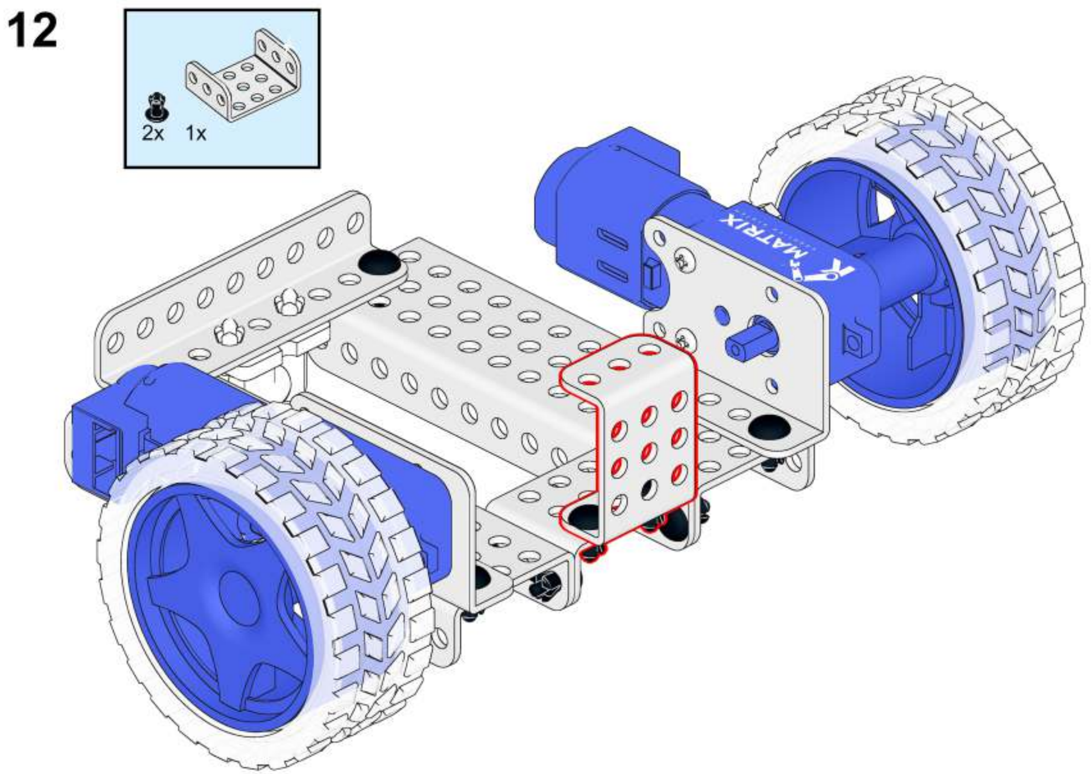
10



Assembly Basic Car



Assembly Basic Car



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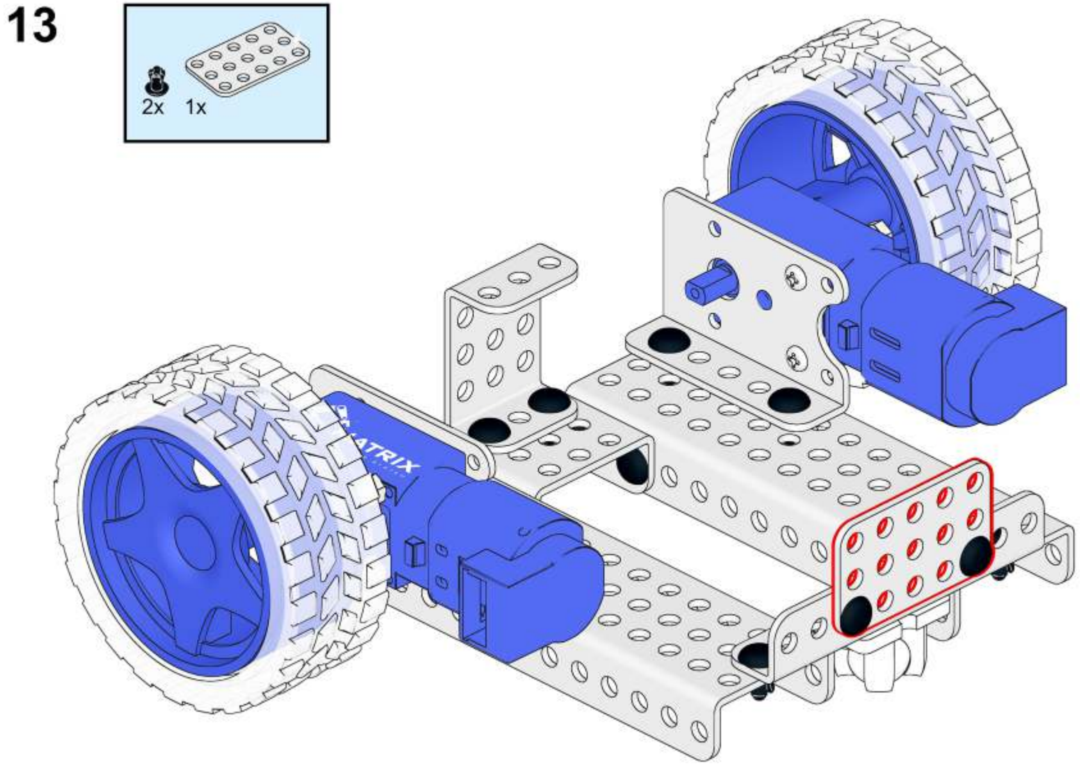

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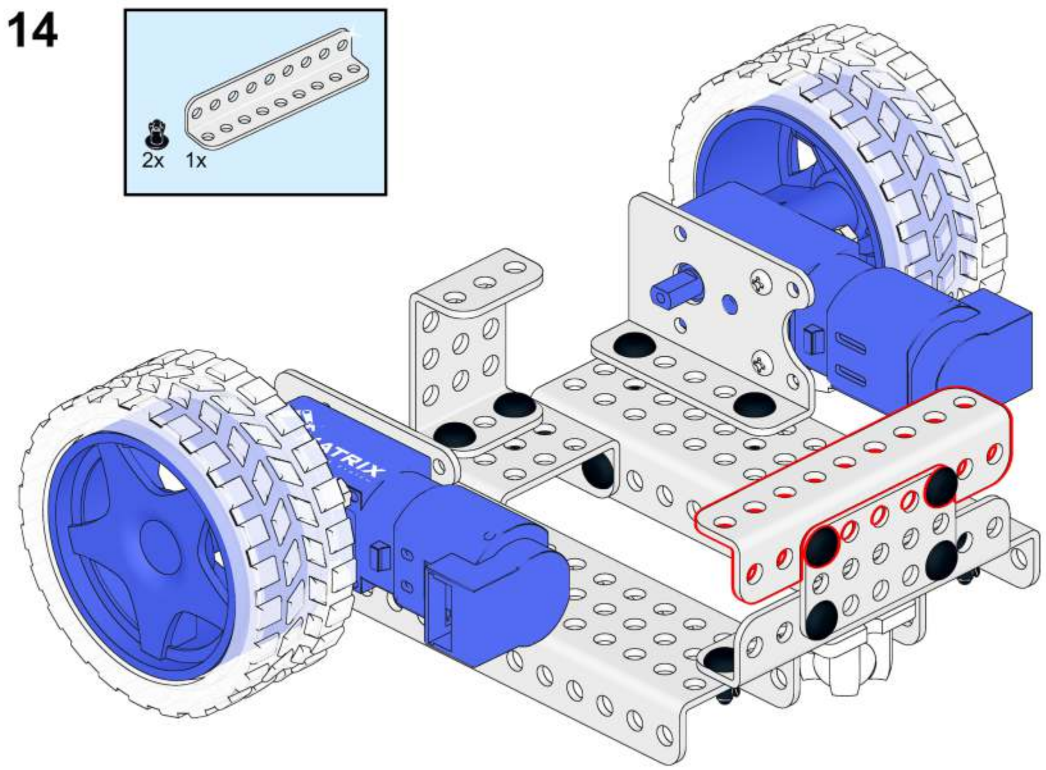

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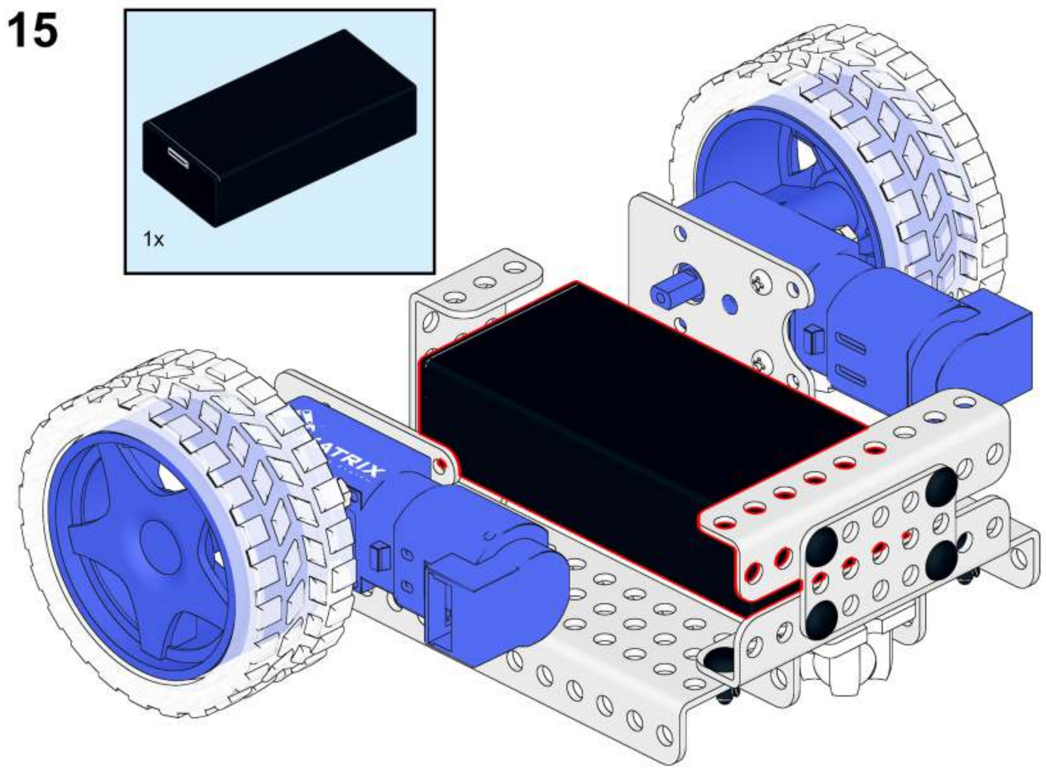

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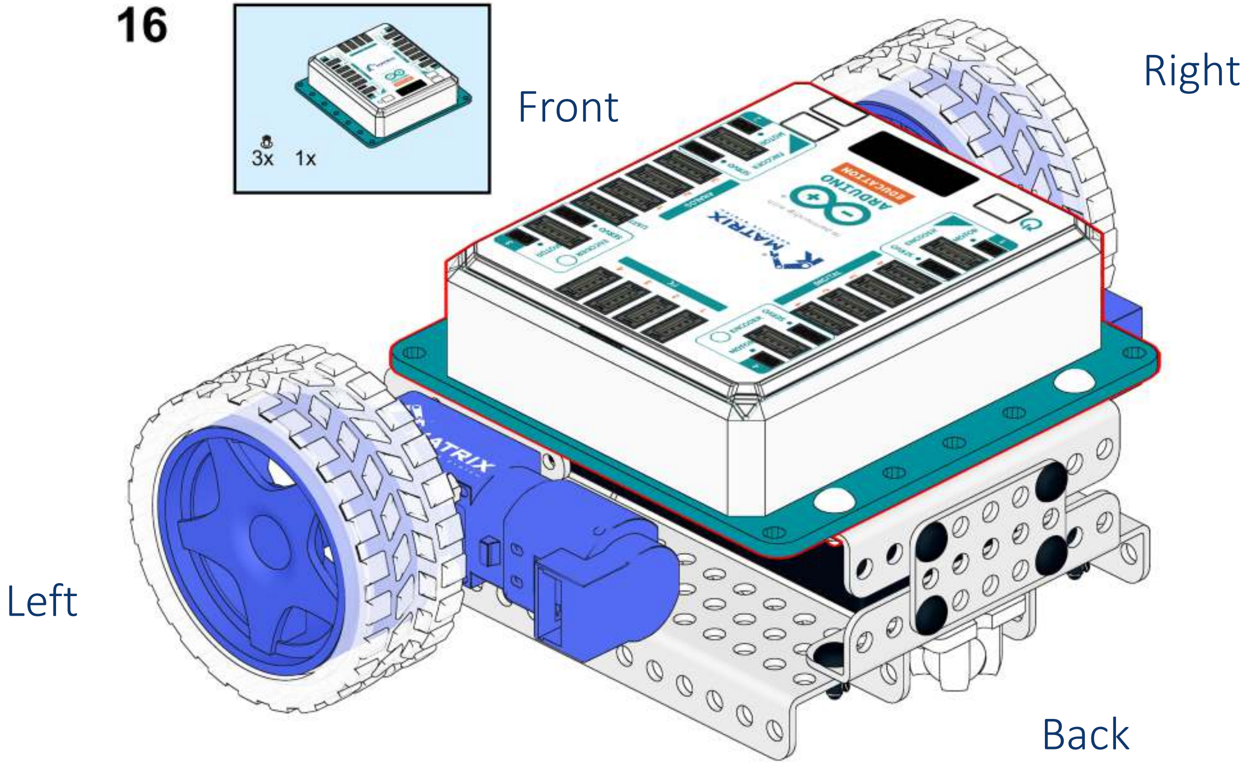
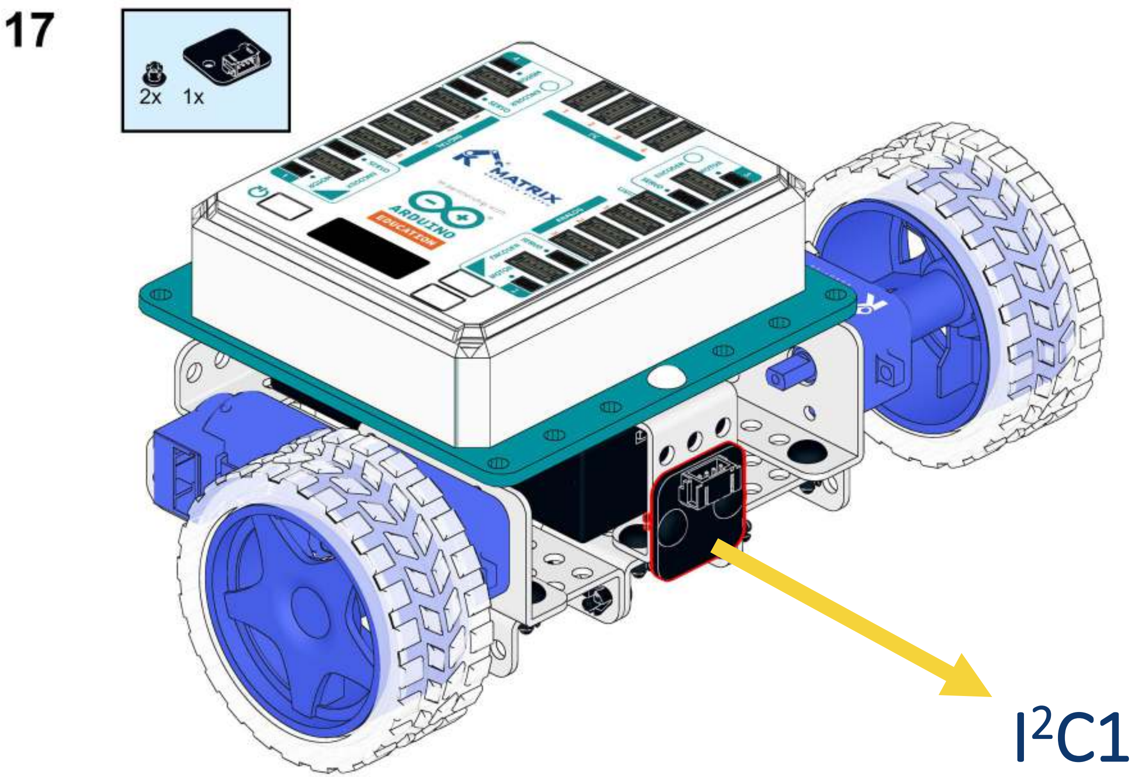


Left motor → Motor 1

Right motor → Motor 2

Black line near green dot “●” .

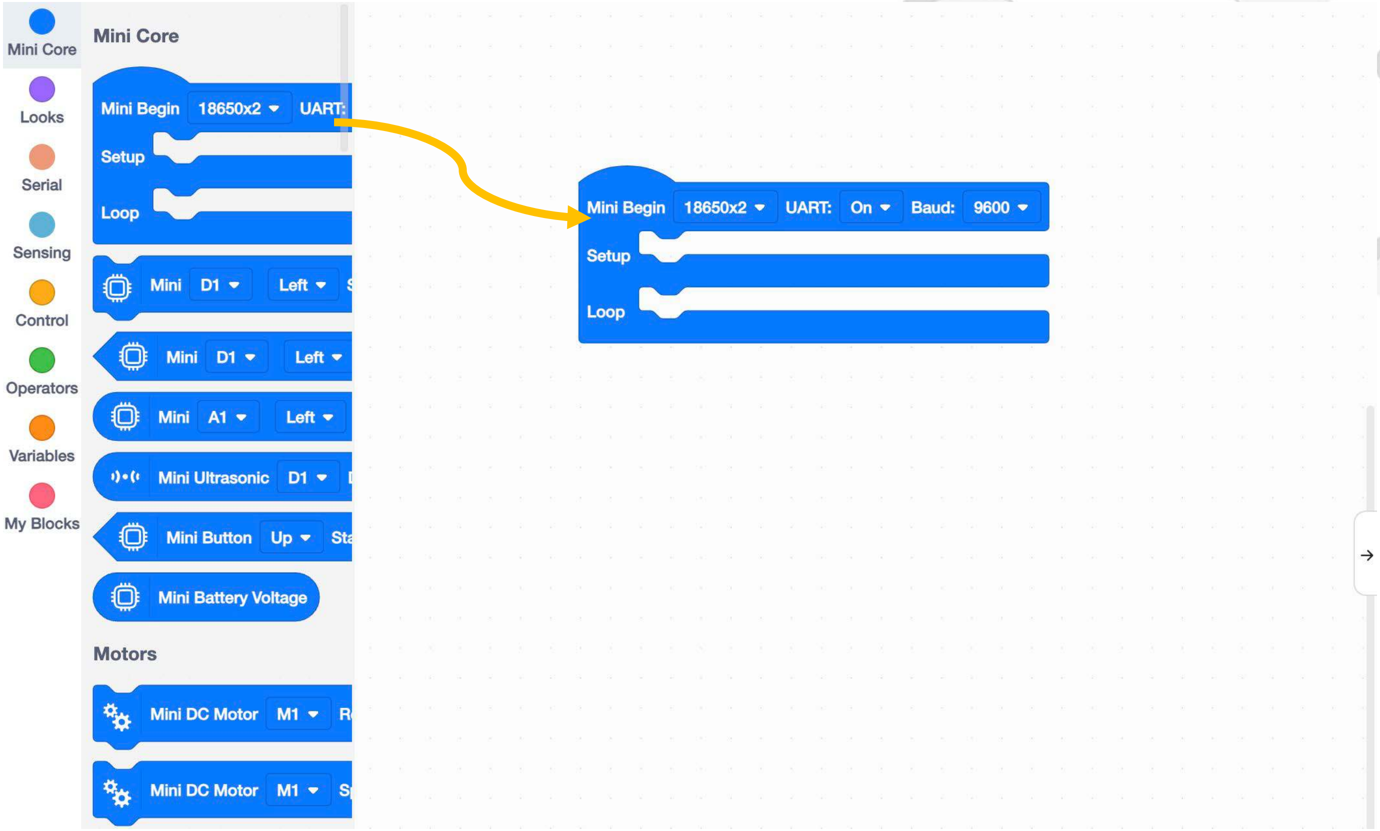
Assembly Laser Sensor



I^2C1

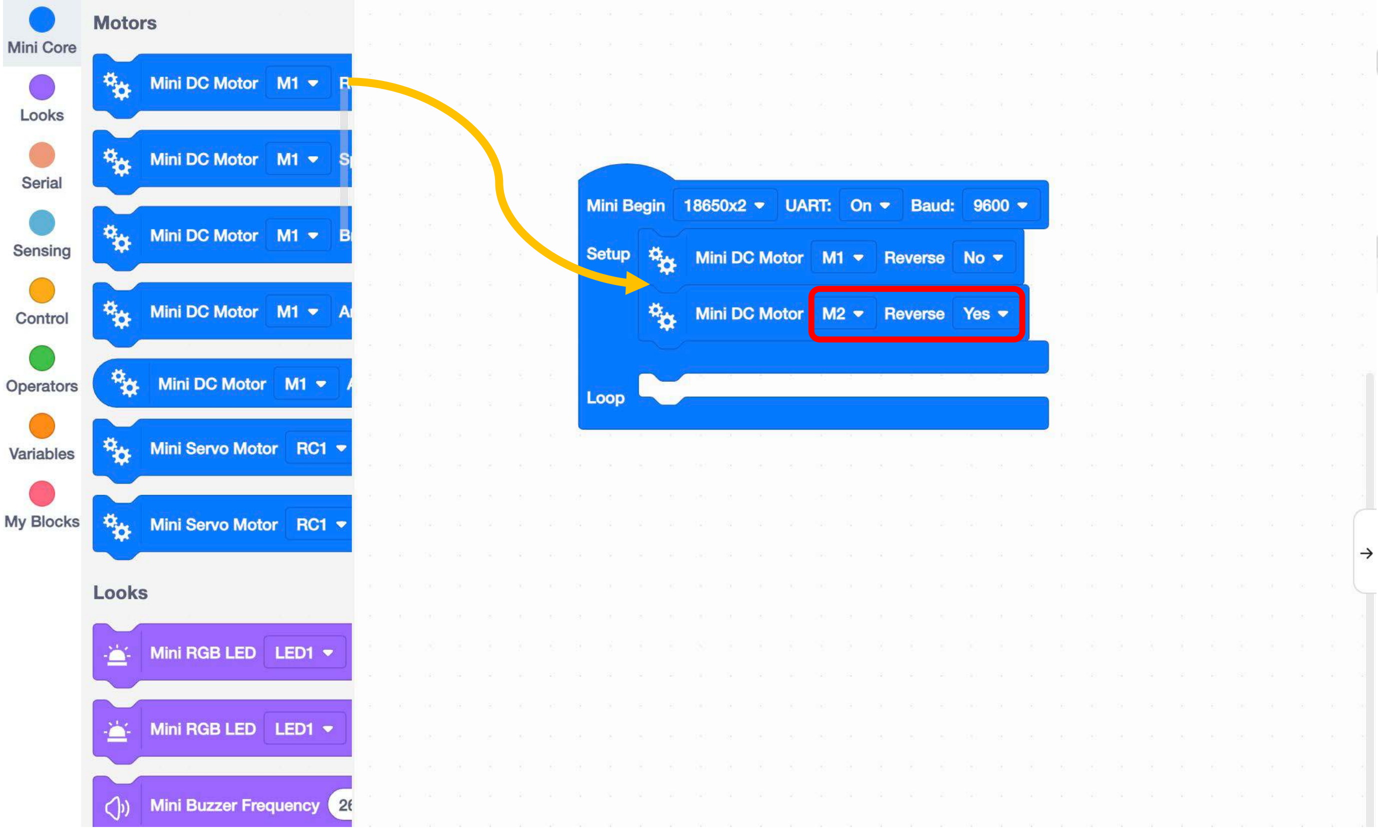
Obstacle Avoiding Robot

1. Drag and drop a Mini Begin block from the Mini Core category into the editing area.



Obstacle Avoiding Robot

2. Drag the 2 Mini DC Motor Reverse blocks to the editing area.

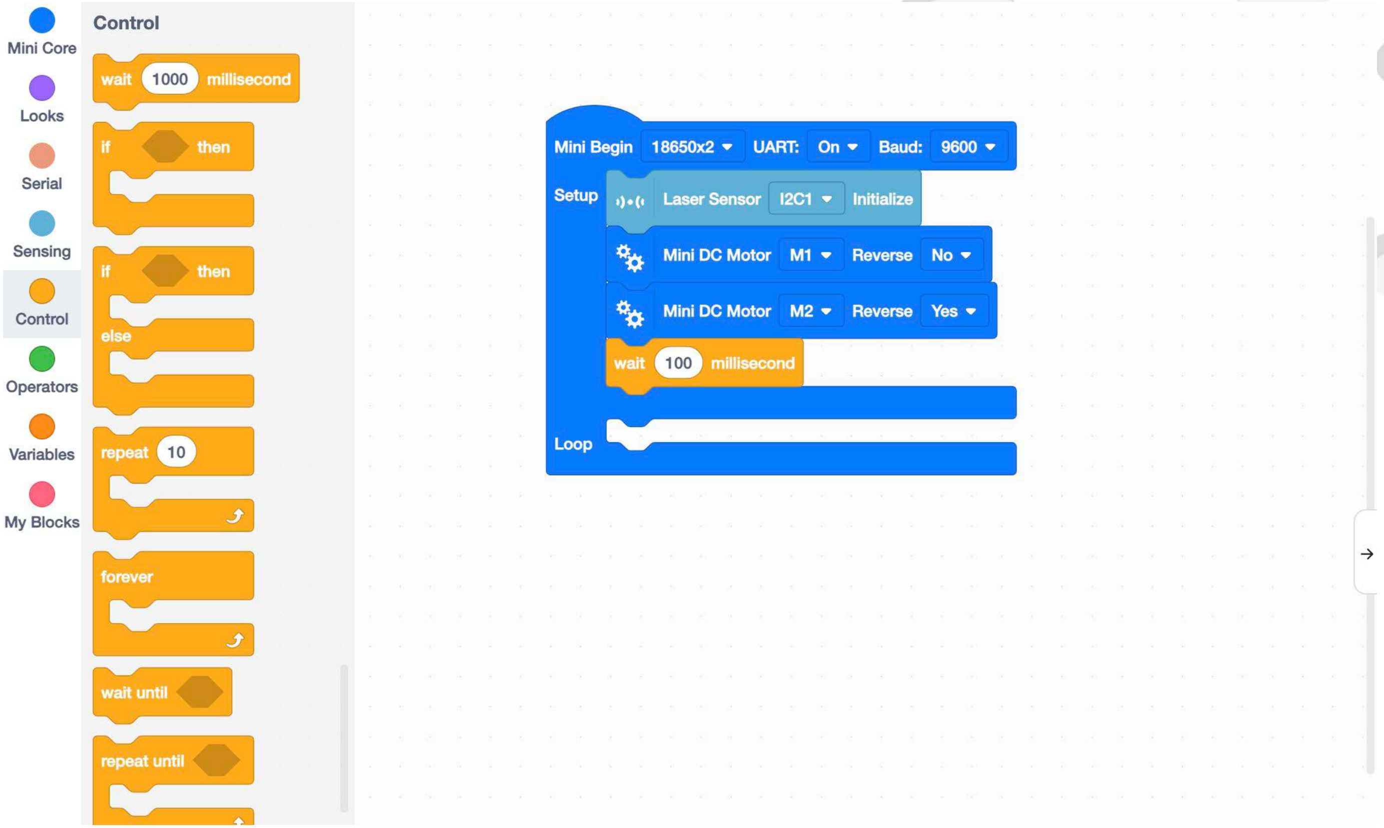


Obstacle Avoiding Robot

3. Drag a Laser Sensor Initialize block to the top of Setup.

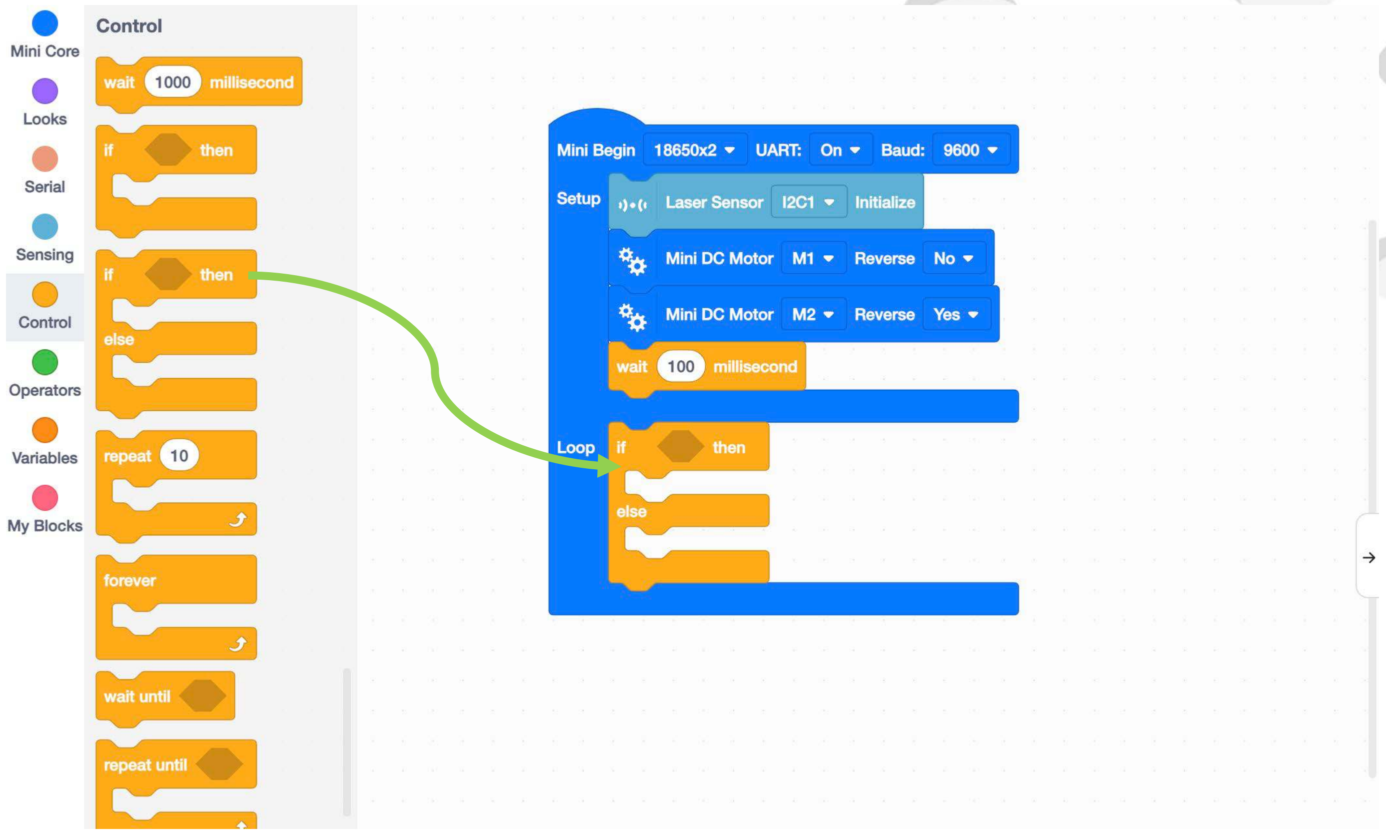
Obstacle Avoiding Robot

4. Insert a delay at the end of Setup.



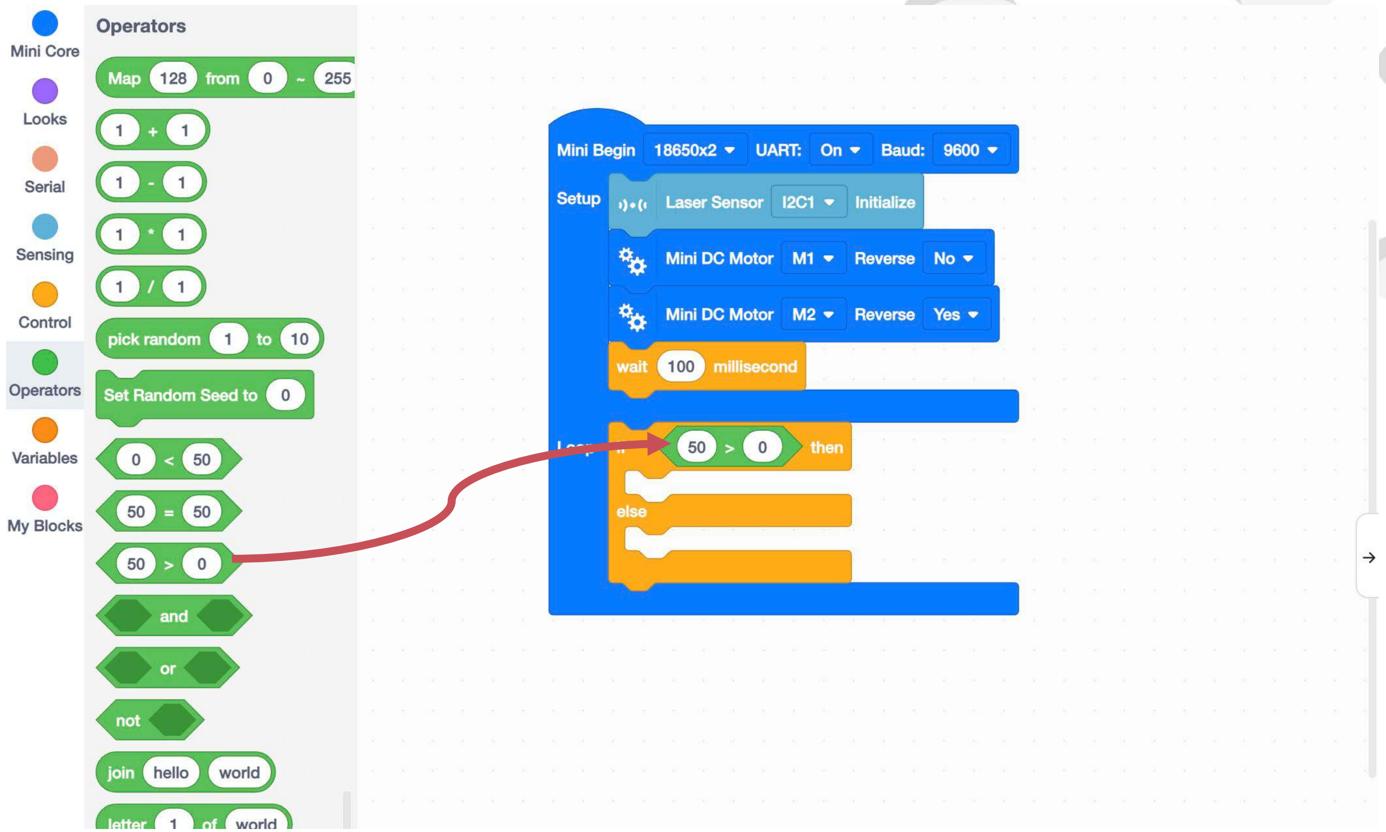
Obstacle Avoiding Robot

5. Drag and drop the "If...then...else..." blocks from the control category to the editing area, and put the block in the Mini initialization loop in order to repeat the condition.



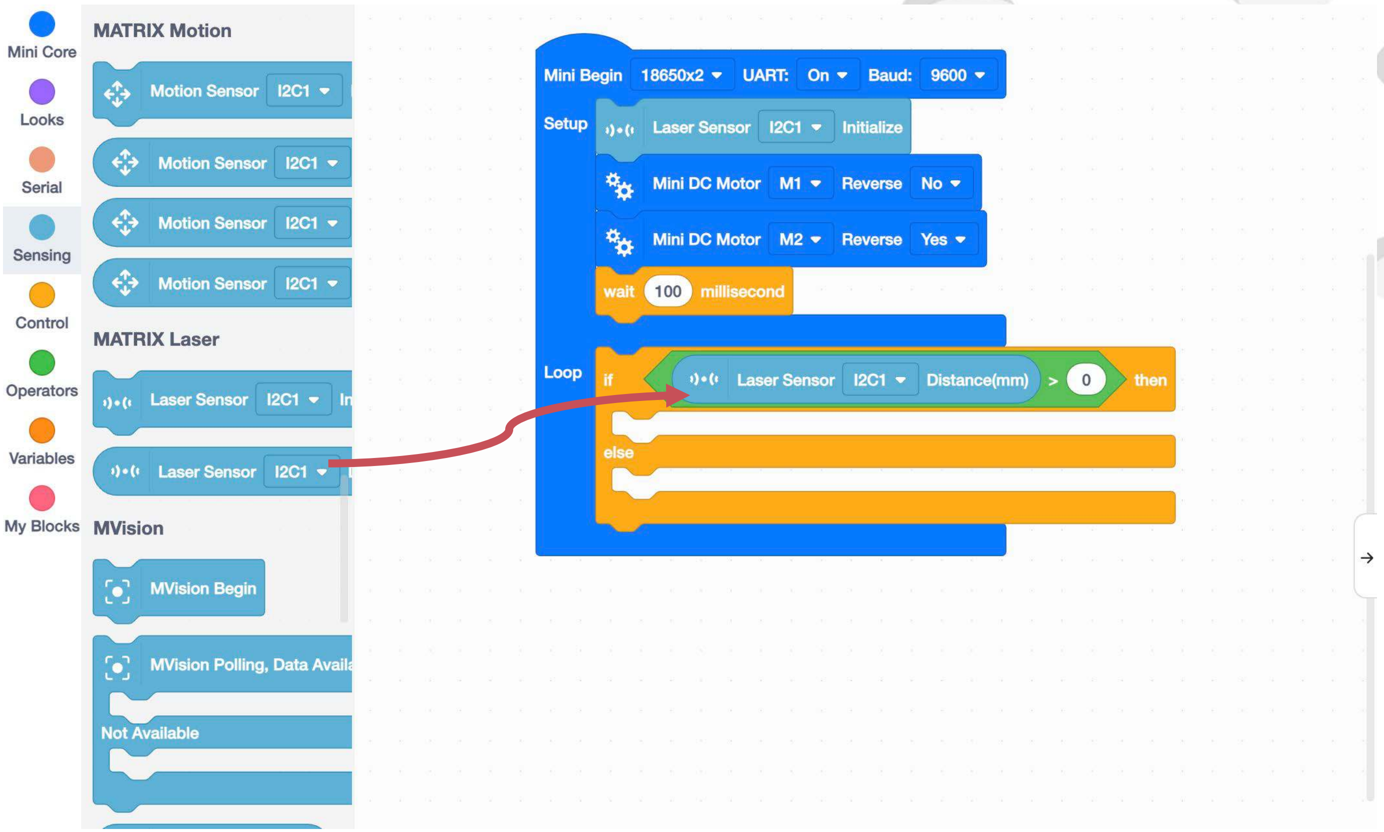
Obstacle Avoiding Robot

6. Drag and drop a "greater than" block from the Operators category between if and then.



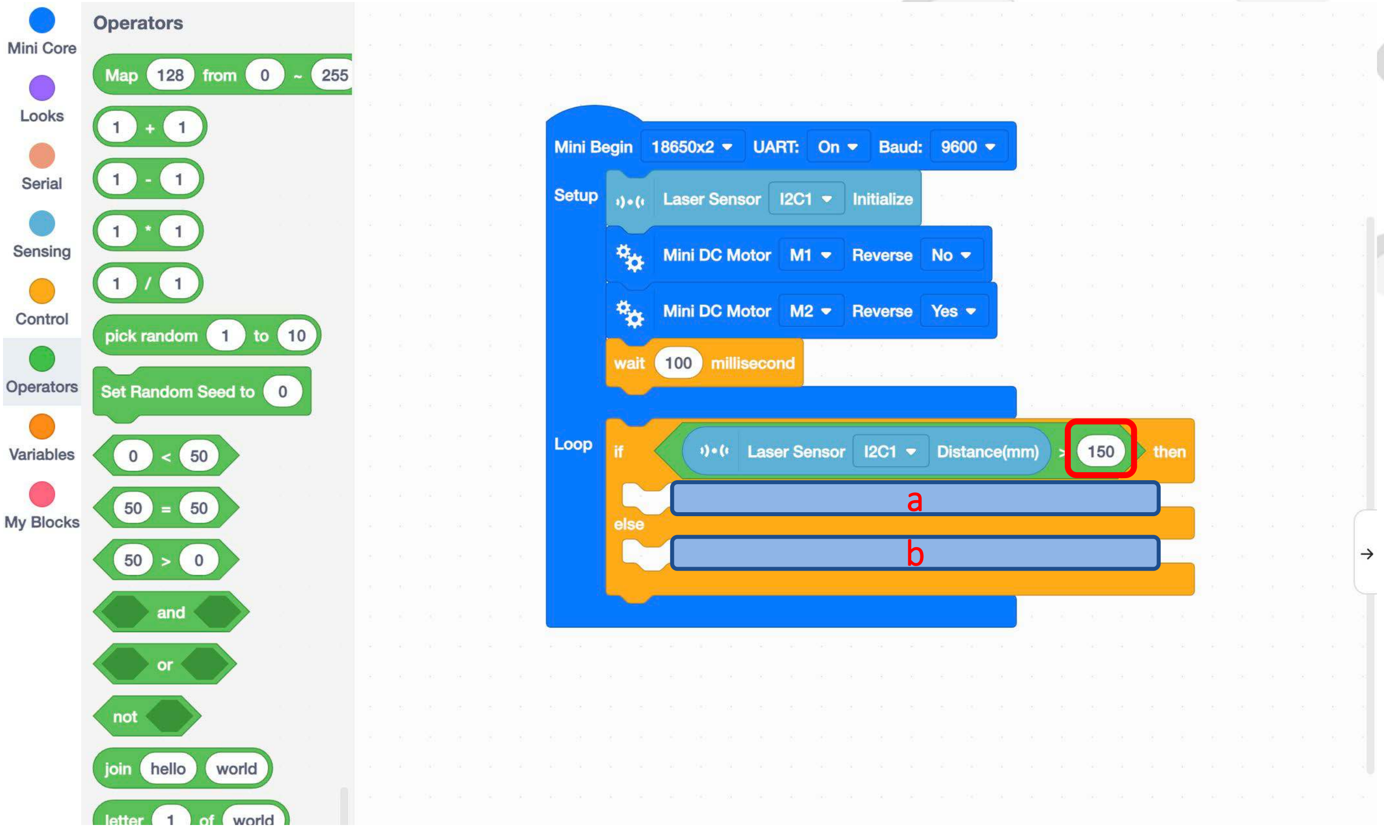
Obstacle Avoiding Robot

7. Drag an MX Laser block from the Sensing category to the first field of the Greater Than block.



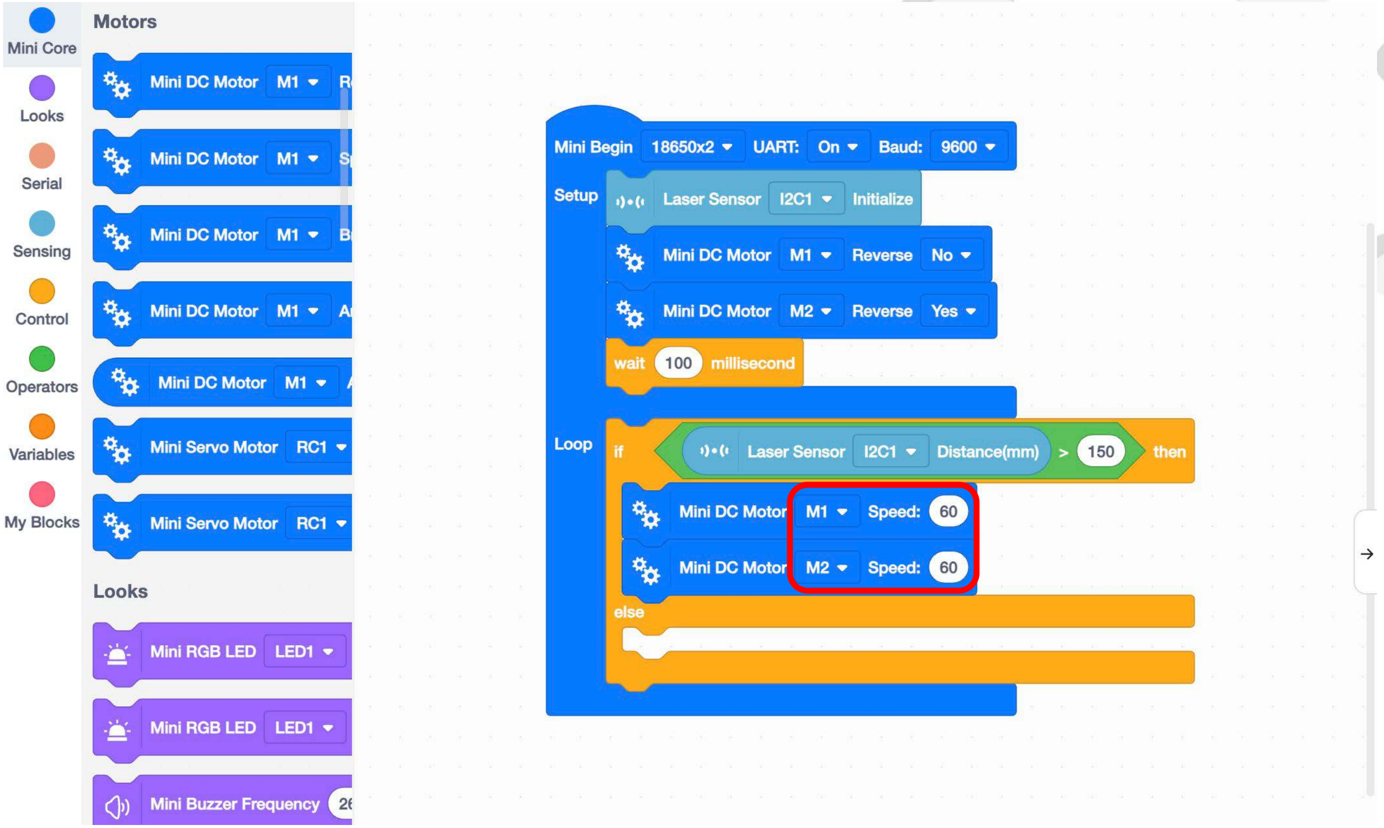
Obstacle Avoiding Robot

8. Enter 150 in the second field of the Greater Than block. The condition is set so that if the distance detected by the laser sensor is greater than 150 mm, then a is executed, otherwise b is executed.



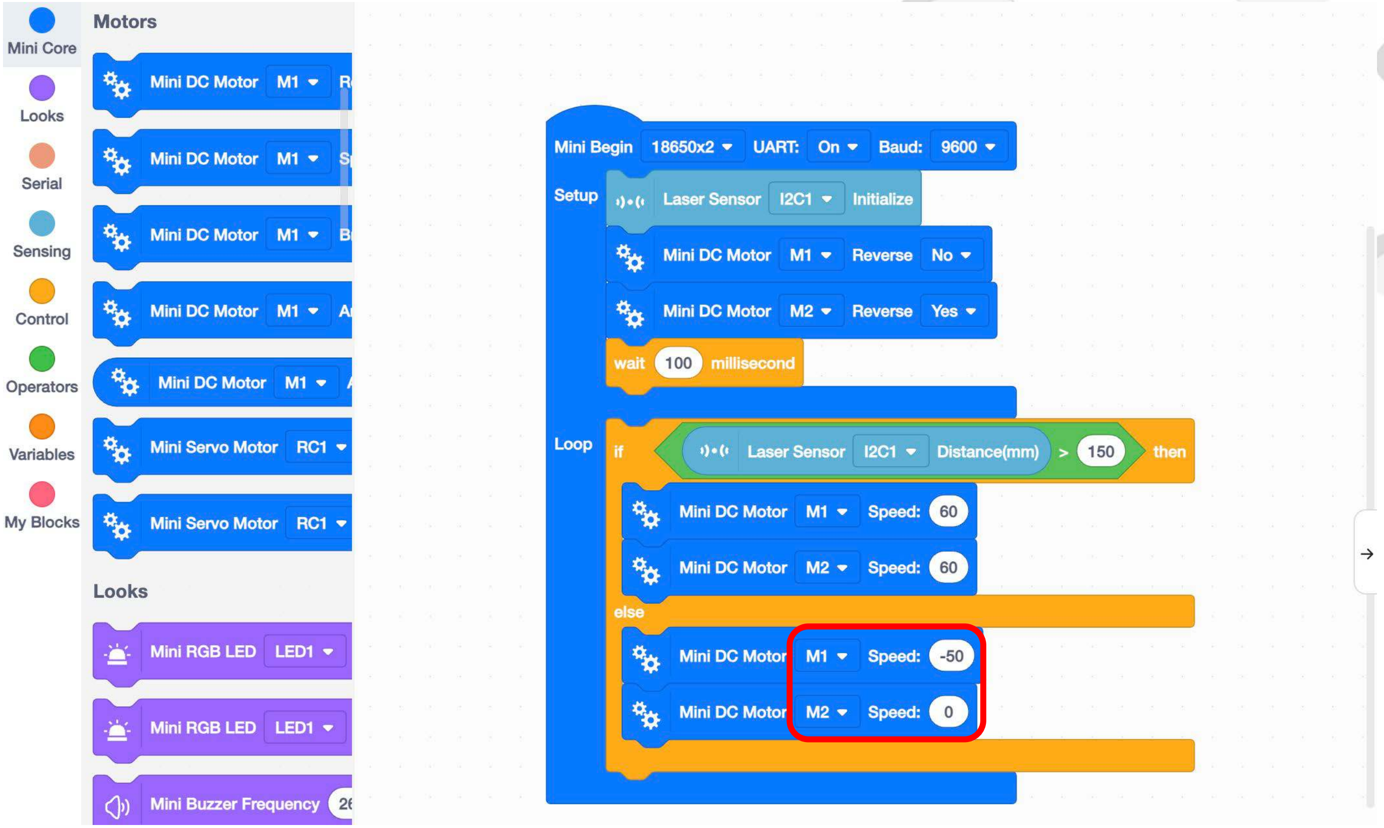
Obstacle Avoiding Robot

9. If the value detected by the laser sensor is greater than 150mm, it means that there is no obstacle in front of the robot, and the robot can go straight.



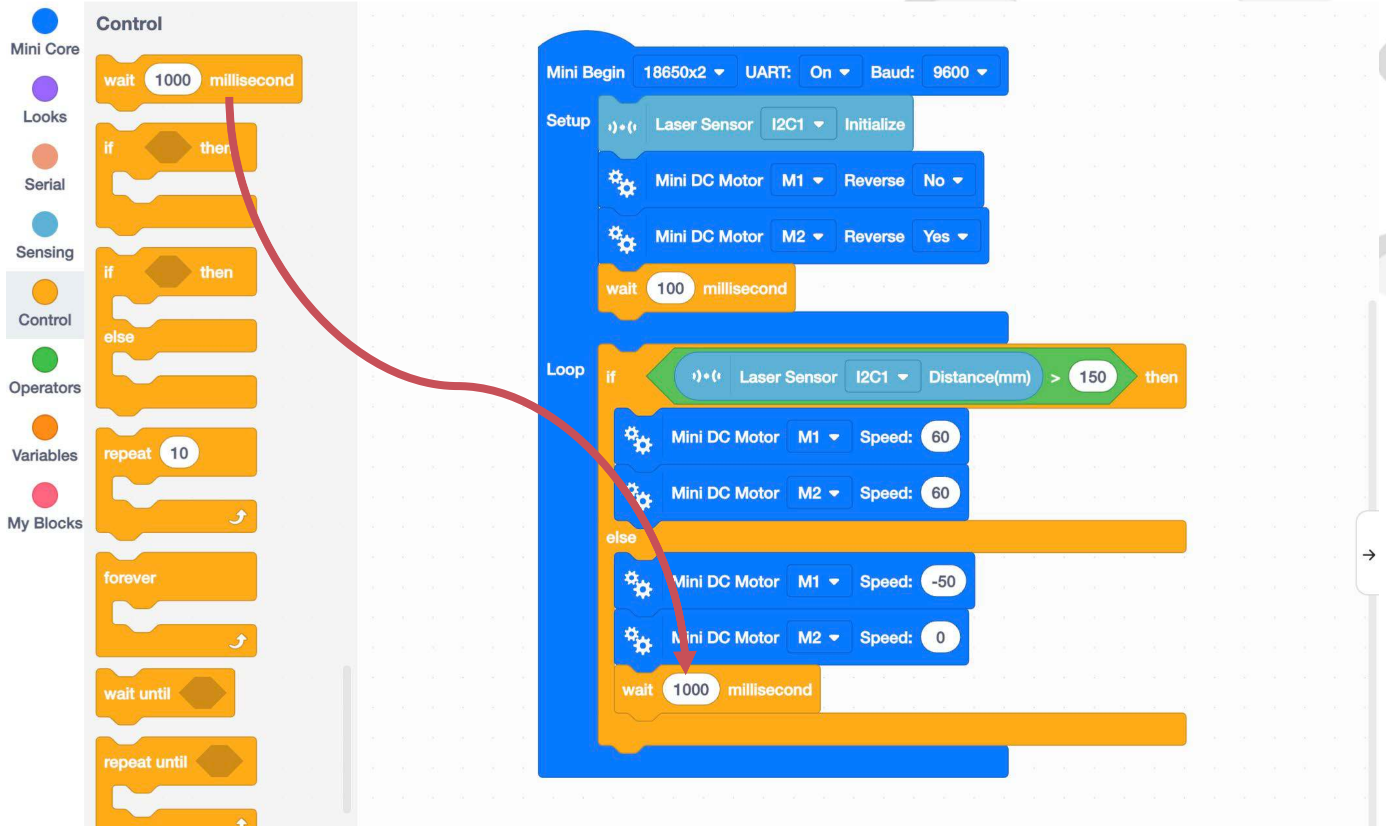
Obstacle Avoiding Robot

10. If the value detected by the laser sensor is not greater than 150mm, it means that there is an obstacle in front of the robot, and the robot need to turn.



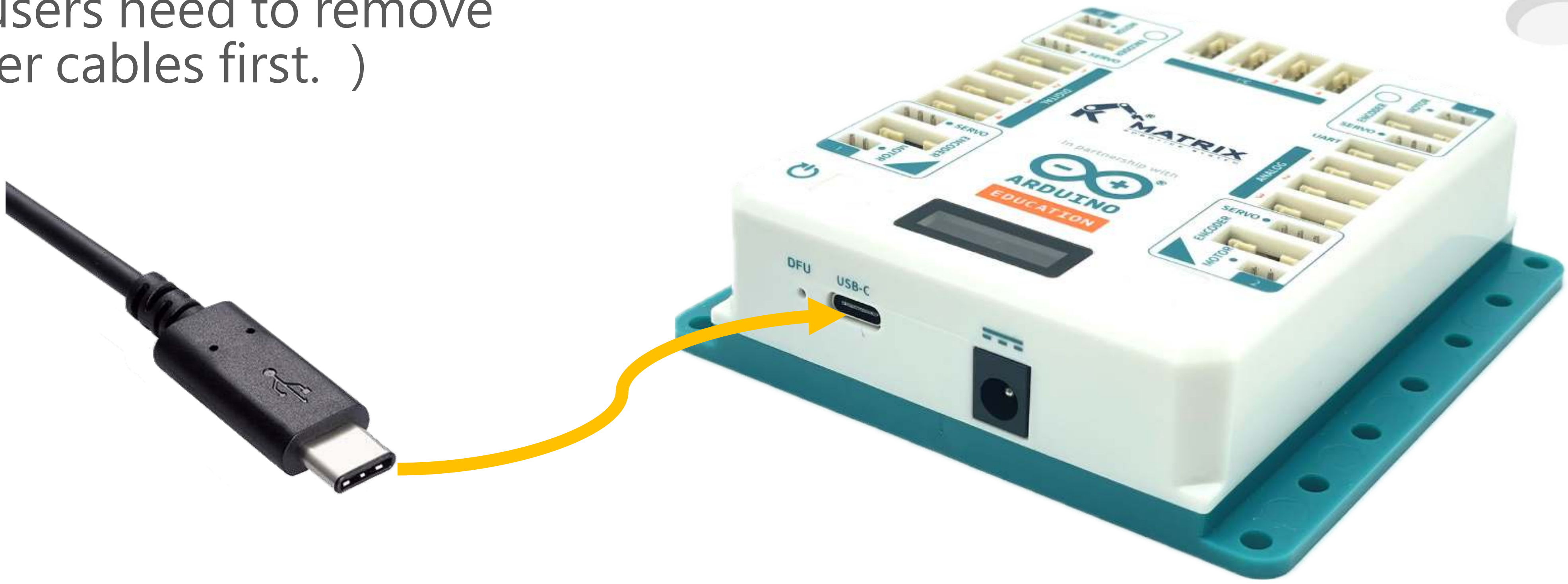
Obstacle Avoiding Robot

11. Set the time.



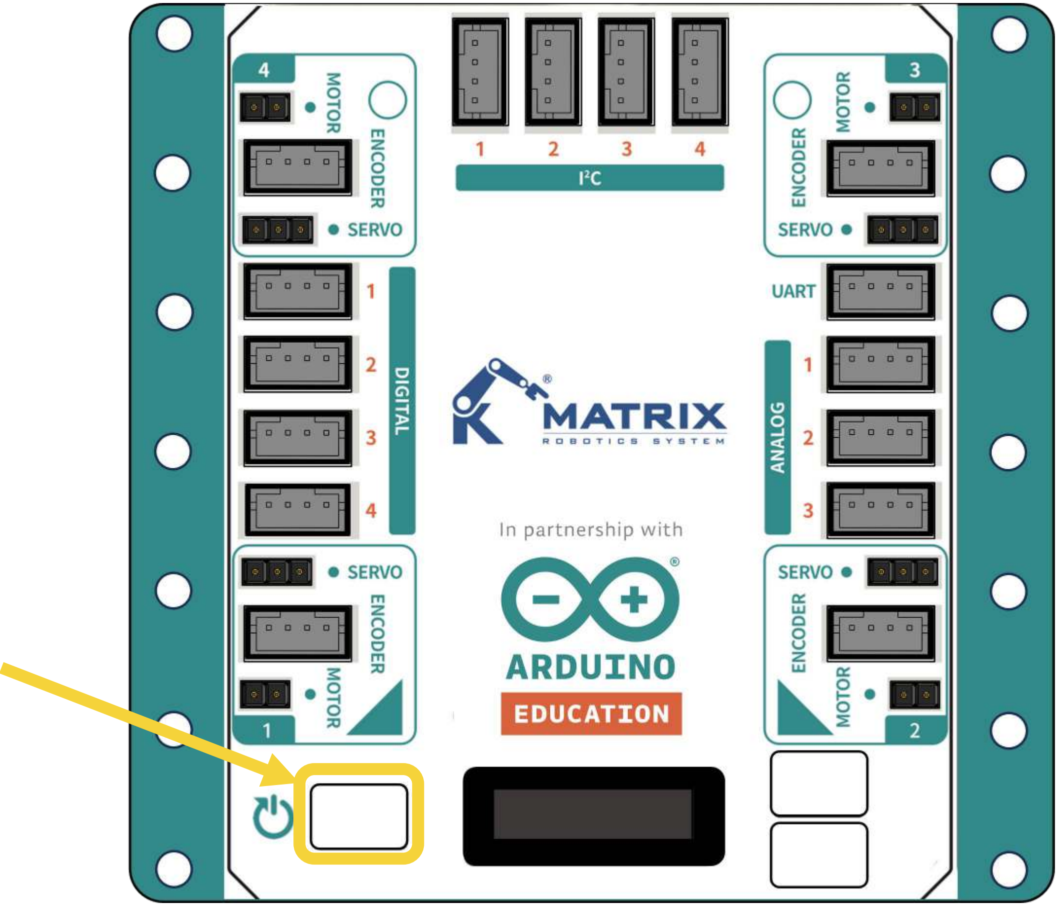
Basic Car-Go Square

12. Connect the controller to the computer using a USB cable.
（ Mac users need to remove the power cables first. ）



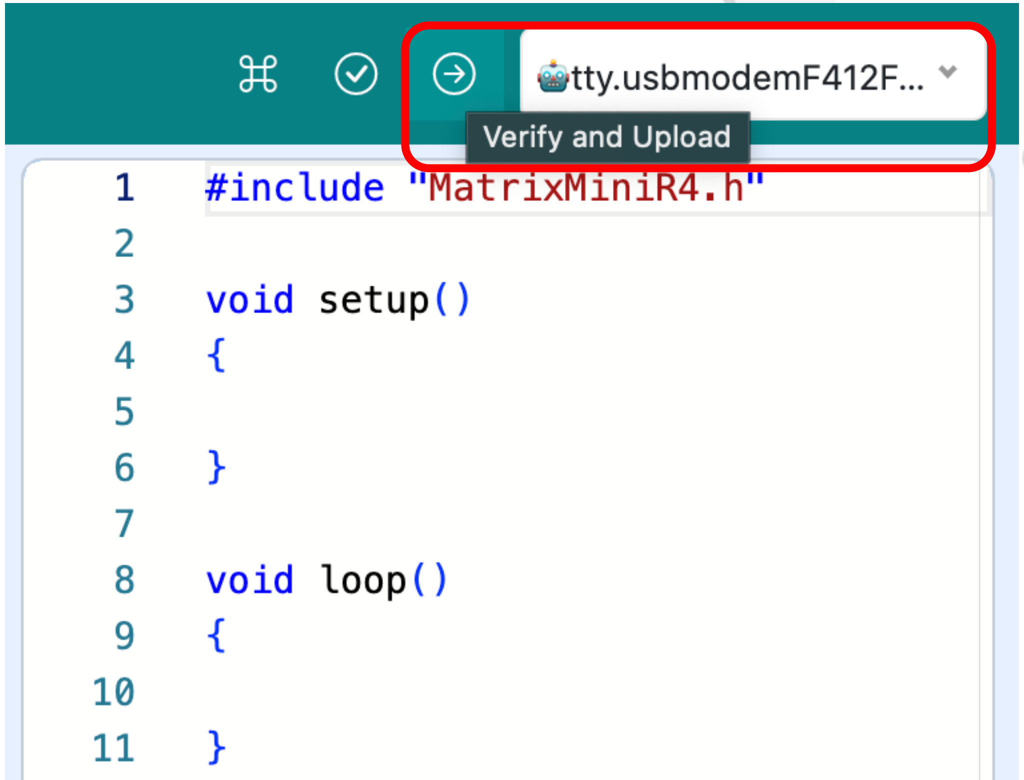
Basic Car-Go Square

13. Press and hold the Reset Button to switch on the Mini R4.



Obstacle Avoiding Robot

14. Select the COM port and upload the program.



Obstacle Avoiding Robot

15. Connect the battery box and press the reset button to run the program.

(Make sure the battery box has switched on)

Congratulations on completing this unit!

Please click on the next unit!